Lec 3 6.1-6.6 8.1-8.6

概念自检

6.1 - 6.6

- Basic Counting Principles
- the Inclusion-Exclusion Principle
- Tree Diagram
- **Pigeonhole Principle**
- **Generalized Pigeonhole Principle**
- Ramsey Number
- Permutation
- Combination
    - Combination corollary
    - Combinatorial proof
- Binomial Theorem
- Pascal's Identity
    - Pascal's triangle
- Vandermonde's Identity
- Permutation with Repetition
- Permutation with Indistinguishable Objects
- r-Circle Permutation
- Conbination with Repitition
    - star-bar model
- Distributing Objects into Boxes
    - four cases
- Generating Permutation
    - lexicographic order
- Generating Combination
- Stirling Number

8.1 - 8.6

- Recurrence Relation
    - a solution of recurrence relation
    - initial conditions
    - degree

    - the Tower of Hanoi
- Linear homogeneous recurrence relation of degree k with constant coefficients
    - Characteristic equation
    - Characteristic root
    - Theorem 1: 二次方程，两个不同的根
    - Theorem 2: 二次方程，重根
    - Theorem 3: k 次方程，k 个不同的根
    - Theorem 4: k 次方程，t 个不同的根，根 r_t 的重数为 m_t
- Linear nonhomogeneous recurrence relation with constant coefficients
    - Theorem 5: 已知某个特定解，可以得出所有一般解
    - Theorem 6: 该特定解的求法
    - > Theorem 1 - Theorem 6 的逻辑其实是很顺的，大家可以配合练习按这个次序试着理一下
- Genrating Function
- Extended binomial coefficient
- Counting eith Generating Functions
- Using Generationg Functions to Solve Recurrence Relations
- the Principle of Inclusion-Exclusion
    - derangement

习题整理

6.1-6.6

1. 25 llf 期中

   Suppose $|A| = 8$ and $|B| = 5$.

   - Find the number of functions $f : A \to B$.

   - How many of these functions are one-to-one?

2. Suppose $|A| = 20$ and $|B| = 24$. The number of one-to-one functions $f : A \to B$ is ?

3. 25 turing quiz4

   Suppose $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. If S has a subset X such that the sum of its elements is odd, we call X an odd subset $(e.g., \{8, 9\})$. If S has a subset Y such that the sum of its elements is even, we call Y an even subset $(e.g., \{2, 4, 8\})$. Which is greater: the number of odd subsets or the number of even subsets? Or are they equal?

第二类斯特林数

4. 25 llf 期中

   ( ) The number of ways to partition a set with 10 elements into 2 nonempty and disjoint subsets is 512.

   注：此题等价于求第二类斯特林数中的 $S(10,2)$

隔板法

5. 25 llf 期中

   How many solutions are there to the equation $x_1+x_2+x_3+x_4=20$ where $x_i, i=1,2,3,4$ is a nonnegative integer such that $x_i \geq 2$ for $i=1,2,3,4$.

6. 25 turing quiz4

   10 balls are arranged in a row from left to right, including 4 indistinguishable red balls and 6 indistinguishable green balls. No two red balls are allowed to be adjacent. How many different arrangements are there?

7. 25 turing quiz4

   10 balls are arranged in a row from left to right, including 4 labeled red balls and 6 labeled green balls. No two red balls are allowed to be adjacent. How many different arrangements are there?

计数综合

8. 19 春夏期末

   How many different ways can you put 9 coins in 9 boxes which are labeled $B_1, \ldots, B_9$ on them

   (1) if the coins are all different and no box is empty?

   (2) if the coins are all different and only two boxes B_1 and B_9 are empty?

(3) if the coins are all different and exactly four boxes are not empty?

(4) if the coins are all different and each box is either empty or contains exactly three coins?

(5) if the coins are identical?

(6) if the coins are identical and exactly six boxes are empty?

9. You have 10 books and 4 boxes. In how many ways can you put the 10

   books into the boxes if

   (1) the books are distinct, and boxes are distinct

   (2) the books are identical, and boxes are distinct

   (3) the books are distinct, and boxes are identical

   (4) the books are identical, and boxes are distinct without lefting empty

鸽巢原理

10. 25 llf 期中

    A professor teaching a Discrete Math course gives a single choice quiz that has ten questions, each with four possible responses: a, b, c, d. What is the minimum number of students that must be in the professor's class in order to guarantee that at least three answer sheets must be identical? (Assume that no answers are left blank.)

11. Prove that every sequence of $n^2 + 1$ distinct integers contains a subsequence of $n + 1$ length that is either strictly increasing or strictly decreasing.

12. An arm wrestler is the champion for a period of 75 hours.(Here, by an hour, we mean a period starting from an exact hour, such as 1 P.M., until the next hour.) The arm wrestler had at least one match an hour, but no more than 125 total matches. Show that there is a period of consecutive hours during which the arm wrestler had exactly 24 matches.

13. 25 turing quiz2

    Let $A = \{a_1, a_2, \ldots, a_8\}$ be a set consisting of 8 distinct positive integers, each satisfying $a_i \leq 35$.

    Prove that A must have two different disjoint subsets whose elements have the same sum.

14. Let x be an irrational number. Show that for some positive integer j not exceeding the positive integer n, the absolute value of the difference between jx and the nearest integer to jx is less than $\frac{1}{n}$.

8.1-8.6

15. llf 25 期中

Find the coefficient of x^{10} in the power series of the function $\frac{x^2}{1-4x^2}$.

16. llf 25 期中

Use generating functions to solve the recurrence relation $a_n = 2a_{n-1} + 2^n + 5$

with initial conditions $a_0 = 1$.

17. llf 25 期中

Given a sequence $\{a_n\}$: $2, 5, 9, 14, 20, 27, 35, \ldots$ (1) Find a recurrence relation for a_n and give the necessary initial condition(s); (2) Find an explicit formula for a_n by solving the recurrence relation in part (1) via its characteristic equation; (3) Write down a generating function in closed form for the sequence $\{a_n\}$.

18. llf 25 期中

How many onto functions are there from a set with *eight* elements to a set with *four* elements?

19. zyw 25 第三次小测

Solve the recurrence relation $a_n = a_{n-1}^2 a_{n-2}^3$ if $a_0 = 2$ and $a_1 = 2$.

20. turing 25 quiz4

How many four-digit decimal numbers ($1000 - 9999$) are divisible by 3 and contain the digit 3?

21. turing 25 quiz4

Solve the following recurrence relation with the given initial conditions:

$$a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3} \quad (n \geq 3),$$

with $a_0 = -5, \quad a_1 = 4, \quad a_2 = 88.$